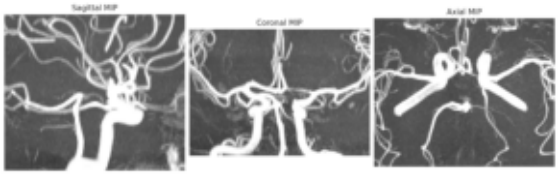
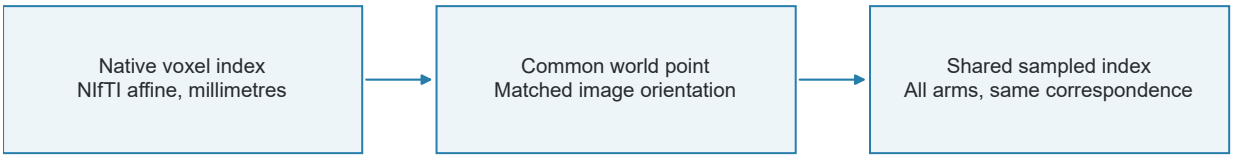
a Native MRA crop



b Input-only vessel surface



c Coordinate and input contract



Only the binary union conditions preprocessing. Class identities remain evaluation targets.

d Sampling does not establish anatomical absence



All examples are fixed DEV illustrations, not augmentation experiments or a clinical segmentation workflow.